

Introduction to Modeling for
Formal Verification

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Today's goals



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Module 1

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Introduction to Model Checking

Motivation of Verification

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Course Updates and Accessibility Support
Reading • 1 min
- Non-Credit Students: Welcome and Where to Find Help
Reading • 10 min
- Motivating Examples
Video • 7 min
- High Profile Previous Bugs and Their Impacts
Video • 10 min
- Principles of Model Checking - Chapter 1, Introduction - Part 1
Reading • 15 min
- Hardware and Software Bugs - Wikipedia
Reading • 20 min
- 🔒

Motivation of Verification Quiz
Graded Assignment • 10 min

Overview of Verification

- Overview of Functional Verification
Video • 5 min
- Techniques for Functional Verification
Video • 10 min
- Functional Verification
Reading • 30 min
- Principles of Model Checking - Chapter 1, Introduction - Part 2
Reading • 15 min
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Overview of Verification Quiz
Graded Assignment • 10 min

Formal Verification and Model Checking

- Overview of Formal Verification and Model Checking
Video • 7 min
- Elements of Model Checking
Video • 6 min
- Challenge to Model Checking: State Space Explosion
Video • 5 min
- Principles of Model Checking - Section 1.1, 1.2
Reading • 30 min
- Formal Verification and Model Checking - Wikipedia
Reading • 30 min
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Formal Verification and Model Checking Quiz
Graded Assignment • 10 min

Module 2

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Modeling

Module 3

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Modeling System Composition

Course Updates and Accessibility Support

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We’re currently making updates to this course to improve accessibility and provide a better learning experience for all learners. As these improvements are rolled out, you may notice small updates or adjustments to course materials. Our goal is to minimize disruption while continuing to improve clarity, usability, and overall course quality.

If you need assistance related to course content accessibility, please reach out to us at ali-support@colorado.edu [🔗](#). We’re happy to help and want to ensure you have the support you need to succeed.

In the meantime, you can continue to use Coursera’s built-in flagging system to share feedback. This includes highlighting items you find helpful as well as flagging anything that might need attention. Your feedback helps our teams better understand what’s working well and where improvements may be needed.

Thank you for learning with us and for your patience and feedback as we continue to improve the course experience.

Go to next item

✔ Completed

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👎 Dislike

🚩 Report an issue

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